Does Money Buy Years? GDP per Capita and Life Expectancy Across Countries, 2000-2015

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Question	Across countries from 2000 to 2015 do countries with higher GDP per capita have a longer life expectancy at birth if I hold adults' mean years of schooling and the year fixed?
Hypothesis	Income buys food and clean water and sanitation and health care so I expect a positive coefficient on log GDP per capita that gets smaller when I hold schooling fixed because schooling raises both income and health.
Headline finding	Holding schooling and the year fixed a country with twice the GDP per capita has a life expectancy 3.41 years higher (95% CI 2.84 to 3.99 years with N = 2,904 country-years in 186 countries). The sign matches my hypothesis and the interval is far from zero but within a country over time the link is less than half as large at 1.52 years per doubling (95% CI 0.55 to 2.49) so this is a conditional association that doesn't show the causal effect of income.
Data	One row is one country-year and the sample has 2,904 of them out of 3,472 World Bank economy-years. Life expectancy and GDP per capita (in constant 2021 PPP dollars) and the government health spending share comes from the World Bank's World Development Indicators and adults' mean years of schooling comes from UNDP's Human Development Report 2025 (all downloaded on 2026-11-07).
Method	OLS of life expectancy on log GDP per capita and mean years of schooling and a dummy for each year with standard errors clustered by country so they allow for unequal variance and for the repeated years of each country.

1. The Question and the Data

Richer countries tend to live longer but what a government or an aid donor should do about it depends on why it happens because if income itself buys longer lives then economic growth is also health policy. But if most of the link comes from schooling or from things that a country can't change like its climate and its history then money spent straight on schools and clinics would do more than waiting for growth, which matters a lot to the finance ministries that set budgets and to the World Bank and the donors that fund them.

My question is whether countries with higher GDP per capita have a longer life expectancy from 2000 to 2015 if I hold adults' schooling and the year fixed, with the outcome being life expectancy at birth in years and the main explanatory variable the natural log of GDP per capita in constant 2021 PPP dollars. I use the log because life expectancy goes up fast with income in poor countries and slows down in rich ones (Section 2). The controls are adults' mean years of schooling and a dummy for each year and my hypothesis is a positive coefficient on log GDP per capita that shrinks once I hold schooling fixed.

Every variable is the result of someone's measurement decisions and the World Bank publishes three of my series without collecting any of them itself. The UN Population Division calculates life expectancy using death records where a country keeps them and surveys and models where it doesn't and it counts the deaths from a war or disaster in the year they happen so some countries have abnormally low life expectancies for certain disastrous years. GDP per capita comes from national accounts that the International Comparison Program's price surveys turn into PPP dollars and WHO collects the health share and UNDP calculates adults' mean years of schooling using censuses and surveys and fills in the years between them.

After the feedback on my checkpoint post I got my numbers for life expectancy and GDP per capita from the World Bank instead of the Kaggle file and I took the health spending share from the World Bank and schooling from UNDP too so every variable comes straight from the agency that publishes it.

Two kinds of places drop out of the World Bank data (Table 1) and the first is the 22 economies outside UNDP's country list like Puerto Rico and Guam that go missing because of their political status which I can see so that part is MAR. The second is 9 of UNDP's 195 countries and apart from tiny rich countries like Liechtenstein and Monaco the data is missing for places like Cuba and North Korea and Yemen because the economy is closed or at war or collapsed so that part ties to the missing income itself and is probably MNAR and neither is MCAR. That leaves out the poorest and most troubled economies so the estimate only concerns countries that publish national accounts and might understate how steep the curve is at the very bottom and Table 2 summarizes the 2,904 country-years in 186 countries that remain.

Table 1. From 3,472 World Bank country-years to the sample

Step	Country-years
Economies x years, World Bank, 2000-2015	3,472
No life expectancy	0
No GDP per capita (PPP)	−341
No mean years of schooling	−227
Country-years in the sample	**2,904**

The steps run in order so a row that's missing two things only counts once at the first step it fails

Table 2. Summary statistics for 2,904 country-years in 186 countries from 2000 to 2015

Variable (units)	Mean	Median	SD	Min	Max	N
Life expectancy at birth (years)	69.36	71.26	9.11	14.66	84.60	2,904
GDP per capita (2021 PPP $)	21,207	12,396	23,285	712	149,025	2,904
Log GDP per capita	9.36	9.43	1.17	6.57	11.91	2,904
Mean years of schooling, adults 25+ (years)	7.98	8.23	3.22	0.98	14.08	2,904
Government health spending (% of government spending)	9.77	9.03	4.88	1.16	37.62	2,862
Year	2007.6	2008	4.60	2000	2015	2,904

Sources: The data come from the World Bank's World Development Indicators (which are based on the UN Population Division and the International Comparison Program and WHO's Global Health Expenditure Database) and from UNDP's Human Development Report 2025 and I downloaded all of it on 2026-11-07 with GDP per capita in constant 2021 international dollars.

2. Descriptive Evidence

Life expectancy is skewed left (skewness −0.70) because 16 countries fall below 50 years in some years and 15 of them are in Sub-Saharan Africa where HIV and war cuts lives short (Figure 1). Table 2 shows the median of 71.3 years beside the mean of 69.4 for that reason and I keep life expectancy in years because the skew is under the self-check's rule of thumb of about 1 and a log would stretch a left skew further.

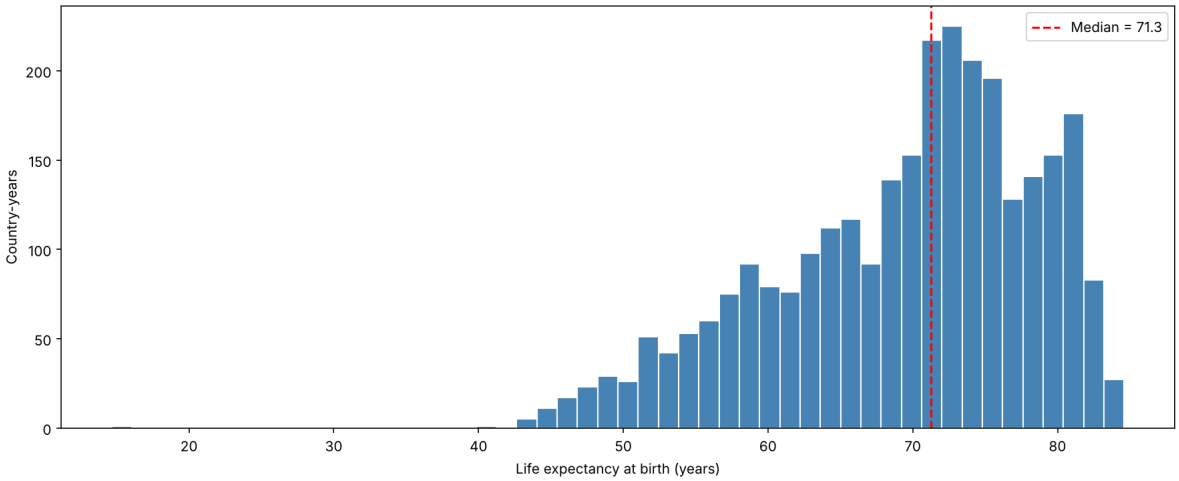


Figure 1. Life expectancy is skewed left with the median country-year (dashed line) at 71.3 years above the mean of 69.4 because a tail of 16 countries falls below 50 in some years (the axis starts so low because of the Central African Republic in 2009). Source: World Bank WDI (UN Population Division).

Figure 2 shows why the model uses log GDP per capita because in dollars (left) life expectancy goes up the fastest among the poorest countries and then barely moves and on a log scale (right) the points sit close to a straight line and the R-squared goes from 0.44 to 0.66.

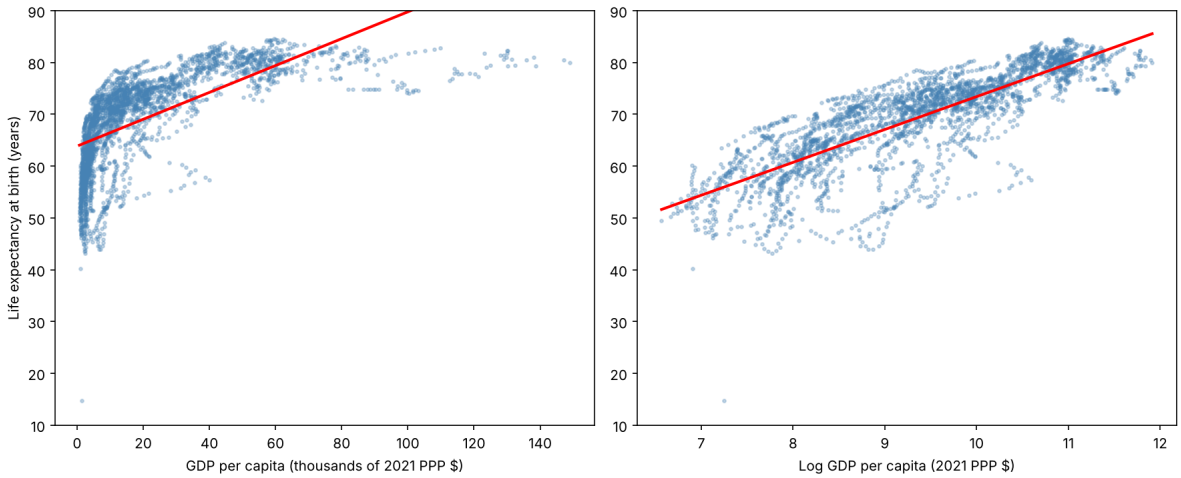


Figure 2. Life expectancy climbs fast with income in poor countries and flattens in rich ones so a straight line misses most countries in dollars (left) but fits the points well in log GDP (right) and the R-squared goes from 0.44 to 0.66. Source: World Bank WDI.

Figure 3 is the reason I don't trust that raw pattern by itself because adults' schooling goes up with both income (correlation 0.76) and life expectancy (0.72) so part of the link in Figure 2 could be schooling.

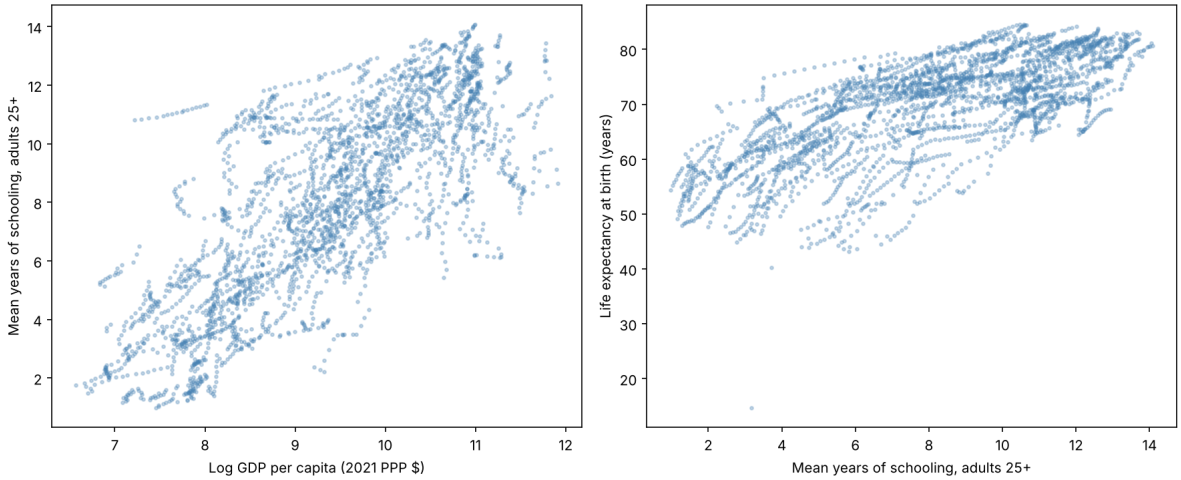


Figure 3. Countries with more schooling are both richer (left with a correlation of 0.76) and live longer (right with a correlation of 0.72) so part of the link between income and life expectancy could be schooling. Source: UNDP HDR 2025 and World Bank WDI.

3. The Model

I fit the model below by OLS with standard errors clustered by country because each country shows up in as many as 16 years of the data.

$$LifeExp_{it} = \beta_0 + \beta_1 \ln(GDPpc_{it}) + \beta_2 Schooling_{it} + \gamma_t + u_{it}$$

Here i is a country and t is a year and γ_t is a dummy for each year and Table A1 in the appendix shows the full results with column 2 as the main model.

Holding schooling and the year fixed the coefficient on log GDP per capita is 4.93 years with a standard error of 0.42 (95% CI 4.10 to 5.75) and doubling income adds ln 2 = 0.693 to its log so a country with twice the GDP per capita of another has a life expectancy 4.93 × 0.693 = 3.41 years higher with a 95% confidence interval from 2.84 to 3.99 years. The interval is far from zero and it's narrow next to the estimate so the data measures the size and sign of the association.

The second coefficient I interpret is schooling and it shows that once I hold GDP per capita and the year fixed a country whose adults have one more year of schooling on average have a life expectancy 0.66 years higher. The standard error is 0.16 and the 95% confidence interval is from 0.35 to 0.96 years so the association is positive and clearly different from zero but it's less exact than the income one.

Without schooling (column 1) the coefficient on log GDP is 6.29 (95% CI 5.75 to 6.83) or 4.36 years per doubling and when I add schooling the coefficient falls to 4.93 and the R^2 rises from 0.67 to 0.69 because part of the raw link comes from schooling.

- Year dummies block a confounding path because new medicine and cheaper vaccines and the spread of HIV treatment raised life expectancy almost everywhere and world growth raised GDP almost everywhere so without year dummies the model would credit income with the progress of the whole world. They would only be a bad control if a country's GDP can change the calendar year and it obviously can't.
- Adults' schooling blocks a confounding path because education raises earnings and also helps people avoid disease (Figure 3) and adults 25 and older finished most of it years before the GDP in my data so income in the same year can't change it much. It would be a bad control if past income paid for that schooling because holding it fixed would take out the part of income's long-run effect that comes from schooling so 3.41 years is the link beyond what comes from schooling and it might understate the full long-run effect of income.
- Government health spending share was kept out of the main model because a richer government can put more of its budget into health so the share sits on the path from income to life expectancy as a mediator and a bad control. Columns 3 and 4 use the same 2,862 rows with and without it and the GDP coefficient only moves from 4.84 to 4.58 so only a small part of the link comes from the health budget share.
- Status (the Kaggle file's developed or developing label) was also left out of the model because it mostly restates income and income decides the label so it would be a bad control just like the health share.
- Country dummies (column 5) block every trait a country keeps over 16 years like its geography and institutions but they change the question to whether a country lives longer when it grows richer than its own average and with them the coefficient falls to 2.19 (95% CI 0.79 to 3.59) or 1.52 years per doubling (95% CI 0.55 to 2.49). I keep column 2 as the main model because my question compares countries and Section 5 comes back to column 5.

4. Diagnostics

I checked the main model with the tests from Topic 14 and the residual plots in Figure A1 show no curve against the fitted values or against log GDP per capita so a straight line in log GDP is enough. The residuals lean negative with a long tail of country-years that live much shorter lives than their income and schooling predict and most of them are in southern Africa.

The spread of the residuals is wide at low and middle incomes and narrow among the richest countries and the Breusch-Pagan test rejects constant variance ($p = 3.07 \times 10^{-9}$) and each country also shows up as many as 16 times so its years aren't separate pieces of evidence so I cluster the standard errors by country to handle both problems. The GDP coefficient's standard error is 0.12 with classical errors and 0.42 when I cluster by country (95% CI 4.10 to 5.75) so clustering more than triples it but that interval is also far from zero.

The variance inflation factors are 2.38 for log GDP per capita and 2.40 for schooling so schooling and income move together (correlation 0.76) without hurting the estimate much because a VIF of 2.4 is low and the interval stays narrow.

Using Cook's distance with the cutoff $4/n = 0.0014$ I find 144 rows in 16 countries that stand out and they're the long left tail from Section 2 led by the Central African Republic in 2009 (0.025). The rest are mostly HIV-hit southern Africa like Botswana and Lesotho and South Africa and oil producers like Equatorial Guinea and Nigeria and Angola whose income is way ahead of their health. Without all 144 rows the coefficient is 4.98 (standard error 0.32 with N = 2,760) against 4.93 with them so they don't affect the result that much.

5. Threats and Conclusion

For the coefficient on log GDP per capita to be the causal effect of income nothing else that effects life expectancy can vary with income once I hold schooling and the year fixed and life expectancy can't feed back into income either. Neither one holds because a tropical climate and a heavy disease load and weak institutions keep a country both poor and short-lived and my data doesn't measure them. If I only compare each country with itself (column 5) then the coefficient falls from 4.93 to 2.19 so more than half of the cross-country link comes from differences between countries that don't change. Healthier workers also earn more so longer lives can raise GDP and wars and disasters and HIV cut lives short in some countries without reducing income first so the influence check flags those same places. Measurement adds noise too because the UN models life expectancy where there are no death records and the sample leaves out closed and war-torn economies like North Korea and Yemen. So this is why my estimate is a conditional association that doesn't show the effect of income and among these 186 countries from 2000 to 2015 a country with twice the GDP per capita has a life expectancy 3.41 years higher with a 95% confidence interval from 2.84 to 3.99 years once I hold adults' schooling and the year fixed.

What I found is that the Preston curve (Preston 1975) holds up in official data because the raw link is strong and schooling explains part of it and the health budget share only a little and what's left has a narrow interval and passes the influence check. A reader should be confident that richer countries live

longer by about this much once schooling is equal but much less confident that raising one country's income would give it the full 3.41 years because within a country the link is less than half as large. With another month I'd use GDP from ten years earlier so health in the same year can't push GDP up and I'd add measures of malaria and HIV and look for income changes that have nothing to do with health like oil price swings.

References

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Appendix

Table A1. Life expectancy regressed on log GDP per capita and controls

	(1) GDP only	(2) Main	(3) Main, health rows	(4) Adds health share	(5) Country dummies
Log GDP per capita	6.29	4.93	4.84	4.58	2.19
	(0.27)	(0.42)	(0.41)	(0.40)	(0.71)
Mean years of schooling (years)		0.66	0.67	0.55	−0.03
		(0.16)	(0.15)	(0.15)	(0.19)
Health spending (% of gov. spending)				0.26	
				(0.08)	
Constant	9.16	16.96	17.78	18.39	40.36
	(2.65)	(3.07)	(2.96)	(2.85)	(5.40)
Country dummies	No	No	No	No	Yes
N	2904	2904	2862	2862	2904
R^2	0.67	0.69	0.70	0.71	0.98

Note. The dependent variable is life expectancy at birth in years and the standard errors in parentheses come from OLS clustered by country with a dummy for each year in every column. Column 2 is the main model and column 3 re-estimates it on the 2,862 rows with a health spending share so column 4 compares the same rows when it adds that share (Section 3). Column 5 adds a dummy for each country so its constant belongs to Afghanistan in 2000.

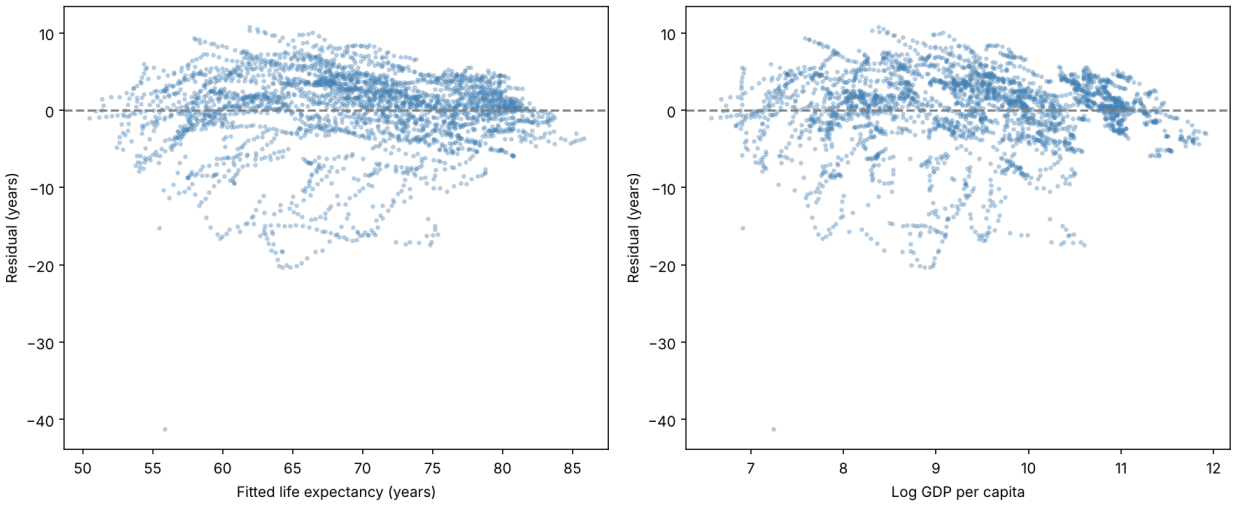


Figure A1. The main model's residuals show no curve against the fitted values (left) or log GDP per capita (right) but they spread wider at low and middle incomes and have a long negative tail that the Central African Republic in 2009 leads.